

Second Order ODEs:

Second order Linear ODE is of the form:

$$y'' + p(x)y' + q(x)y = r(x)$$

It is homogeneous if $r(x) = 0$

Ex: • $y'' + 25y = e^{-x} \cos x$: Linear, Non-homogeneous

• $xy'' + y' + xy = 0$

$\Rightarrow y'' + \frac{1}{x}y' + y = 0$: Linear, homogeneous

• $y''y + (y')^2 = 0$: Non-linear

Principle of Superposition:

The linear combination of any two solutions of a second order linear homogeneous ODE is again a solution. That is, if y_1 and y_2 are solutions then $c_1y_1 + c_2y_2$ is also a solution for any constants c_1 and c_2 .

Proof: Given: $y_1'' + p(x)y_1' + q(x)y_1 = 0$
and $y_2'' + p(x)y_2' + q(x)y_2 = 0$

$$\text{Then } (c_1 y_1 + c_2 y_2)'' + p(x) (c_1 y_1 + c_2 y_2)' + q(x) (c_1 y_1 + c_2 y_2)$$

$$= c_1 y_1'' + c_2 y_2'' + p(x) (c_1 y_1' + c_2 y_2') + q(x) (c_1 y_1 + c_2 y_2)$$

$$= c_1 (y_1'' + p(x) y_1' + q(x) y_1) + c_2 (y_2'' + p(x) y_2' + q(x) y_2)$$

$$= c_1 \cdot 0 + c_2 \cdot 0 = 0$$

Ex: $y'' + y = 0$

$y_1 = \cos x$, $y_2 = \sin x$ are solutions $y_1' = -\sin x$, $y_2' = \cos x$

Then $y = c_1 \cos x + c_2 \sin x$

is also a solution.

$y_1'' = -\cos x$, $y_2'' = -\sin x$

$y_1'' + y_1 = 0$, $y_2'' + y_2 = 0$

Ex: The above doesn't work with non-linear ODEs.

$$y'' y - x y' = 0$$

$y_1 = x^2$, $y_2 = 1$

$y_2' = 0$, $y_2'' = 0$

$y_1'' = 2$, $y_1' = 2x$

$0 \cdot y - x \cdot 0 = 0$

$y_1'' y_1 - x y_1' = 2x^2 - x \cdot 2x = 2x^2 - 2x^2 = 0$

Let $y = y_1 + y_2 = 1 + x^2$

$y' = 2x$, $y'' = 2$

Then $y'' y - x y' = 2(1 + x^2) - x(2x) = 2 + 2x^2 - 2x^2$

$\Rightarrow y'' y - x y' = 2 \neq 0$

Ex: The above doesn't work with nonhomogeneous ODE:

$$y'' + y = 1$$

$$y_1 = 1 + \cos x, \quad y_2 = 1 + \sin x$$

$$y_1' = -\sin x$$

$$y_2' = \cos x, \quad y_2'' = -\sin x$$

$$y_1'' = -\cos x$$

$$y_2'' + y_2 = -\sin x + 1 + \sin x$$

$$y_1'' + y_1 = -\cos x + 1 + \cos x = 1$$

$$= 1$$

$$y = y_1 + y_2 = 2 + \cos x + \sin x$$

$$y' = -\sin x + \cos x, \quad y'' = -\cos x - \sin x$$

$$\text{Now } y'' + y = -\cos x - \sin x + 2 + \cos x + \sin x = 2 \neq 1$$

So, $y = y_1 + y_2$ is not a solution.

Basis or Fundamental System of Solutions:

For a second order linear homogeneous ODE, two solutions y_1 and y_2 are said to form a basis or fundamental system of solutions if

y_1 and y_2 are linearly independent

$$\text{i.e. } c_1 y_1 + c_2 y_2 = 0 \Rightarrow c_1 = c_2 = 0$$

In other words, y_1 and y_2 shouldn't be a multiple of each other OR $\frac{y_1}{y_2}$ shouldn't be a constant.

General Solution: If y_1 and y_2 form a basis of solutions for a second order linear homogeneous ODE then $y(x) = c_1 y_1(x) + c_2 y_2(x)$ is the

general solution.

We get a particular solution by taking specific values of c_1 and c_2 .

Ex: $y'' + y = 0$

$y_1 = \cos x$, $y_2 = \sin x$ are solutions of the ODE

$$\frac{y_1}{y_2} = \frac{\cos x}{\sin x} = \cot x \rightarrow \text{not a constant}$$

$\Rightarrow y_1$ and y_2 are linearly independent

$\Rightarrow y(x) = c_1 \cos x + c_2 \sin x$ is the general solution.

Initial Value Problem (IVP):

$$\left. \begin{aligned} y'' + p(x)y' + q(x)y &= 0 \\ y(x_0) &= y_0 \\ y'(x_0) &= m_0 \end{aligned} \right\}$$

Ex: $y'' + y = 0$, $y(0) = 3$, $y'(0) = -0.5$

The general solution is $y(x) = c_1 \cos x + c_2 \sin x$

Then $y'(x) = -c_1 \sin x + c_2 \cos x$

$$y(0) = 3 \Rightarrow c_1 = 3$$

$$y'(0) = -0.5 \Rightarrow c_2 = -0.5$$

Hence the particular solution is:
 $y(x) = 3 \cos x - 0.5 \sin x$

Ex: $(1-x^2)y'' - 2xy' + 2y = 0$

Note that $y_1 = x$ is a solution of the above ODE.

$(y_1' = 1, y_1'' = 0; \text{ substituting LHS} = 0 - 2x(1) + 2x = 0 = \text{RHS})$

Let us try to find another solution $y_2 = u y_1$ where u is a non-constant function (ie. $\frac{y_2}{y_1} = u$)

Now $y_2' = u' y_1 + u y_1' = u' x + u$

$y_2'' = u'' x + 2u'$

Then $(1-x^2)(u'' x + 2u') - 2x(u' x + u) + 2u x = 0$

$\Rightarrow x(1-x^2)u'' + 2(1-x^2)u' - 2x^2u' - 2u x + 2u x = 0$

$\Rightarrow x(1-x^2)u'' + 2(1-x^2)u' - 2x^2u' = 0$

$\Rightarrow u'' + \frac{2}{x}u' - \frac{2x}{1-x^2}u' = 0$

Let $u' = V$ Then $\frac{dV}{V} = -\frac{2}{x}dx + \frac{2x}{1-x^2}dx$ (Reduction of order)

$\Rightarrow \int \frac{dV}{V} = -\int \frac{2}{x} dx + \int \frac{2x}{1-x^2} dx$

$\Rightarrow \ln|V| = -2\ln|x| - \ln|1-x^2|$

$\Rightarrow \ln|V| = -\ln(|x|^2 |1-x^2|)$

$\Rightarrow V = \frac{1}{x^2(1-x^2)}$

$\Rightarrow V = \frac{1}{x^2} + \frac{1}{1-x^2}$

$\Rightarrow V = \frac{1}{x^2} - \frac{1}{(x-1)(x+1)}$

(ignoring the constant as we want a particular solution)

$$\Rightarrow V = \frac{1}{x^2} - \frac{1}{2} \left[\frac{1}{x-1} - \frac{1}{x+1} \right]$$

$$\Rightarrow u' = \frac{du}{dx} = \frac{1}{x^2} - \frac{1}{2} \left[\frac{1}{x-1} - \frac{1}{x+1} \right]$$

$$\Rightarrow \int du = \int \frac{1}{x^2} dx + \frac{1}{2} \int \left[\frac{1}{x+1} - \frac{1}{x-1} \right] dx$$

$$\Rightarrow u = -\frac{1}{x} + \frac{1}{2} \ln \left| \frac{x+1}{x-1} \right| \quad \left(\text{ignoring the constant} \right)$$

$$\Rightarrow y_2 = u y_1 = u x = \left[-1 + \frac{1}{2} x \ln \left| \frac{x+1}{x-1} \right| \right]$$

So, the general solution is

$$y = c_1 y_1 + c_2 y_2 = \left[c_1 x + c_2 \left(-1 + \frac{1}{2} x \ln \left| \frac{x+1}{x-1} \right| \right) \right]$$

where c_1 and c_2 are constants

(Ref : Legendre function)

Ex: $(x^2-x)y'' - xy' + y = 0$

Clearly $y_1 = x$ is a solution since $y_1' = 1$, $y_1'' = 0$
and $(x^2-x) \times 0 - x \times 1 + x = 0$

Let us try to find another solution $y_2 = u y_1$
where u is a non-constant function.

$$\left(\text{i.e. } \frac{y_2}{y_1} = u \right)$$

$$\text{Now } y_2' = u y_1' + u' y_1 = u'x + u$$

$$y_2'' = u''x + u' + u' = u''x + 2u'$$

$$\text{Then } (x^2-x)(u''x + 2u') - x(u'x + u) + ux = 0$$

$$\left(\begin{array}{l} \text{Reduction} \\ \text{of order} \end{array} \right) \Rightarrow (x^2-x)xu'' + (2x^2-2x-x^2)u' + (-x+x)u = 0$$

$$\Rightarrow x^2(x-1)u'' + (x^2-2x)u' = 0$$

$$\Rightarrow x^2(x-1)u'' + x(x-2)u' = 0$$

$$\Rightarrow x(x-1)u'' + (x-2)u' = 0$$

$$\text{Let } u' = v \quad \text{Then } x(x-1)v' + (x-2)v = 0$$

$$\Rightarrow \frac{dv}{v} = - \frac{(x-2)dx}{x(x-1)}$$

$$\Rightarrow \frac{dv}{v} = \left[\frac{1}{x-1} - \frac{2}{x} \right] dx$$

$$\begin{aligned} \Rightarrow \ln v &= \ln(x-1) - 2\ln x & \left(\begin{array}{l} \text{ignore the} \\ \text{constant of} \\ \text{integration} \end{array} \right) \\ \Rightarrow \ln v &= \ln \left(\frac{x-1}{x^2} \right) \\ \Rightarrow u' = v &= \frac{x-1}{x^2} \\ \Rightarrow u' &= \frac{1}{x} - \frac{1}{x^2} \end{aligned}$$

$$\Rightarrow u = \ln|x| + \frac{1}{x}$$

$$\text{So, } y_2 = uy_1 = x(\ln|x| + \frac{1}{x}) = x\ln|x| + 1$$

Thus $y_1 = x$ and $y_2 = x\ln|x| + 1$ and they are linearly independent. Hence they form a basis of solutions of the ODE and the general solution is: $y = c_1 y_1 + c_2 y_2 = c_1 x + c_2 (x\ln|x| + 1)$

Note: $-\frac{(x-2)}{x(x-1)} = \frac{A}{x} + \frac{B}{x-1}$

$$\Rightarrow -\frac{(x-2)}{x(x-1)} = \frac{Ax - A + Bx}{x(x-1)}$$

$$\Rightarrow \frac{-x+2}{x(x-1)} = \frac{(A+B)x - A}{x(x-1)}$$

$$\Rightarrow A+B = -1, A = -2 \Rightarrow B = 1, A = -2$$

$$\text{Hence } -\frac{(x-2)}{x(x-1)} = \frac{1}{x-1} - \frac{2}{x}$$

Reduction of Order:

Consider the ODE: $y'' + p(x)y' + q(x)y = 0$.

Let y_1 be a solution of it.

Design the second solution as: $y_2 = u y_1$

$$\text{Then } y_2' = u' y_1 + u y_1'$$

$$y_2'' = u'' y_1 + u' y_1' + u' y_1' + u y_1''$$

$$\Rightarrow y_2'' = u'' y_1 + 2u' y_1' + u y_1''$$

$$\text{Then } (u'' y_1 + 2u' y_1' + u y_1'') + p(x) (u' y_1 + u y_1') + q(x) u y_1 = 0$$

$$\Rightarrow u'' y_1 + (2y_1' + p(x) y_1) u' + \underbrace{(y_1'' + p(x) y_1' + q(x) y_1)}_{=0} u = 0$$

$$\Rightarrow u'' y_1 + (2y_1' + p(x) y_1) u' = 0$$

$$\text{Let } u' = v$$

$$\Rightarrow y_1 v' + (2y_1' + p(x) y_1) v = 0$$

$$\Rightarrow \frac{dv}{v} = - \frac{2y_1' + p(x) y_1}{y_1} dx$$

$$\Rightarrow \frac{dv}{v} = - \left(\frac{2y_1'}{y_1} + p(x) \right) dx$$

$$\Rightarrow \ln|v| = -2\ln|y_1| - \int p(x) dx$$

$$\Rightarrow v = \frac{1}{y_1^2} e^{-\int p(x) dx}$$

$$\text{Thus } u' = \frac{1}{y_1^2} e^{-\int p(x) dx}$$

$$\Rightarrow u = \int \frac{1}{y_1^2} e^{-\int p(x) dx} dx$$

$$\text{Hence } y_2 = u y_1 = \boxed{y_1 \int \frac{1}{y_1^2} e^{-\int p(x) dx} dx}$$

Second Order Homogeneous ODEs with constant coefficients:

First we look at $y' + ky = 0$

and try if $y = e^{\lambda x}$ is a solution.

$$\text{Then } \lambda e^{\lambda x} + k e^{\lambda x} = 0$$

$$\Rightarrow (\lambda + k) e^{\lambda x} = 0$$

$$\Rightarrow \lambda + k = 0 \Rightarrow \lambda = -k \rightarrow \lambda_1$$

$$\text{Then } y_1 = e^{\lambda_1 x} = e^{-kx}$$

$$\text{General solution: } y = c_1 y_1 = c_1 e^{-kx}$$

Now we consider $y'' + ay' + by = 0$ where $a, b \in \mathbb{R}$

let us try if $y = e^{\lambda x}$ is a solution.

$$y'' = \lambda^2 e^{\lambda x}, \quad y' = \lambda e^{\lambda x}$$

Then

$$\begin{aligned} & (\lambda^2 + a\lambda + b) e^{\lambda x} = 0 \\ \Rightarrow & \boxed{\lambda^2 + a\lambda + b = 0} \rightarrow \boxed{\text{Characteristic Equation}} \end{aligned}$$

Case 1: $a^2 - 4b > 0$ and so there are two

distinct real roots of the characteristic equation

Let the roots be λ_1 and λ_2 ($\lambda_1 \neq \lambda_2$)

Then $e^{\lambda_1 x}$ and $e^{\lambda_2 x}$ are two linearly independent solutions $\left(\frac{e^{\lambda_2 x}}{e^{\lambda_1 x}} = e^{(\lambda_2 - \lambda_1)x} \neq \text{Constant} \right)$
since $\lambda_2 - \lambda_1 \neq 0$

So, the general solution is

$$\begin{aligned} y &= c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x} \\ &= c_1 e^{\frac{1}{2}(-a + \sqrt{a^2 - 4b})x} + c_2 e^{\frac{1}{2}(-a - \sqrt{a^2 - 4b})x} \end{aligned}$$

Case 2: $a^2 - 4b = 0$ and so the real roots are equal
(a double root)

So the root is $\frac{-a \pm \sqrt{a^2 - 4b}}{2} = -\frac{a}{2} = \lambda$ (say)

So, $y_1 = e^{\lambda x} = e^{-\frac{a}{2}x}$ is one solution

Then by reduction of order, the second solution is given by $y_2(x) = e^{\lambda x} \int \frac{1}{(e^{\lambda x})^2} e^{-\int a dx} dx$

$$\Rightarrow y_2(x) = e^{-\frac{a}{2}x} \int \frac{1}{\left(e^{-\frac{a}{2}x}\right)^2} e^{-ax} dx$$

$$= e^{-\frac{a}{2}x} \int \frac{e^{-ax}}{e^{-ax}} dx = e^{-\frac{a}{2}x} x = x e^{-\frac{a}{2}x}$$

So, the general solution is

$$y(x) = c_1 y_1 + c_2 y_2 = c_1 e^{\lambda x} + c_2 x e^{\lambda x}$$

$$= e^{\lambda x} (c_1 + c_2 x) = \boxed{e^{-\frac{a}{2}x} (c_1 + c_2 x)}$$

Case 3: $a^2 - 4b < 0$ and so the roots are complex.

The two roots are $\lambda_1 = \frac{-a + \sqrt{a^2 - 4b}}{2} = -\frac{a}{2} + i\omega$

and $\lambda_2 = \frac{-a - \sqrt{a^2 - 4b}}{2} = -\frac{a}{2} - i\omega$

where $\omega = \frac{\sqrt{4b - a^2}}{2} \in \mathbb{R}$

(hence $\frac{\sqrt{a^2 - 4b}}{2} = i\omega$)

So, $y_1 = e^{\lambda_1 x} = e^{(-\frac{a}{2} + i\omega)x} = e^{-\frac{a}{2}x} (e^{i\omega x})$

$$\Rightarrow y_1 = e^{-\frac{a}{2}x} (\cos(\omega x) + i \sin(\omega x))$$

and $y_2 = e^{\lambda_2 x} = e^{(-\frac{a}{2} - i\omega)x} = e^{-\frac{a}{2}x} (e^{-i\omega x})$

$$\Rightarrow y_2 = e^{-\frac{a}{2}x} (\cos(\omega x) - i \sin(\omega x))$$

Now by using principle of superposition we can say $Y_1 = \frac{1}{2} Y_1 + \frac{1}{2} Y_2$ and $Y_2 = \frac{1}{2i} Y_1 - \frac{1}{2i} Y_2$ are also solutions.

So, $Y_1 = \frac{1}{2} Y_1 + \frac{1}{2} Y_2 = e^{-\frac{a}{2}x} \cos(\omega x)$ is a solution
and $Y_2 = \frac{1}{2i} Y_1 - \frac{1}{2i} Y_2 = e^{-\frac{a}{2}x} \sin(\omega x)$ is another solution.

Also $\frac{Y_1}{Y_2} = \cot(\omega x)$ is not a constant and

hence Y_1 and Y_2 are linearly independent solutions.

$$\begin{aligned} \text{Therefore } Y(x) &= c_1 e^{-\frac{a}{2}x} \cos(\omega x) + c_2 e^{-\frac{a}{2}x} \sin(\omega x) \\ &= e^{-\frac{a}{2}x} (c_1 \cos(\omega x) + c_2 \sin(\omega x)) \end{aligned}$$

is the general solution.

Ex: $Y'' - 0.7Y + 0.12Y = 0$

The characteristic equation is: $\lambda^2 - 0.7\lambda + 0.12 = 0$
 $\Rightarrow (\lambda - 0.4)(\lambda - 0.3) = 0$

It has roots 0.4 and 0.3.

The corresponding general solution is:

$$Y = c_1 e^{0.4x} + c_2 e^{0.3x}$$

Ex: $x^2 y'' - 5x y' + 9y = 0 \quad (x > 0) \quad (\Rightarrow y'' - \frac{5}{x} y' + \frac{9}{x^2} y = 0)$

Now $y_1 = x^3$ is a solution of the above equation

Since $x^2 y_1'' - 5x y_1' + 9y_1 = x^2 \times 6x - 5x \times 3x^2 + 9x^3$
 $= 6x^3 - 15x^3 + 9x^3 = 0$

Now using method of reduction of order

$$u(x) = \int \frac{1}{x^6} e^{-\int -\frac{5}{x} dx} dx = \int \frac{1}{x^6} e^{5 \ln|x|} dx$$

$$= \int \frac{1}{x^6} e^{5 \ln x} dx = \int \frac{x^5}{x^6} dx = \int \frac{1}{x} dx = \ln x$$

So, $y_2 = u(x) y_1 = (\ln x) x^3$ is another solution.

Hence the general solution is $y(x) = c_1 x^3 + c_2 (\ln x) x^3$

Ex: $y'' - 2y' - 3y = 0$, $y(0) = 2$, $y'(0) = 14$

Characteristic Equation is: $\lambda^2 - 2\lambda - 3 = 0$

$$\Rightarrow (\lambda - 3)(\lambda + 1) = 0$$

$$\Rightarrow \lambda = 3, -1$$

Hence the general solution is $y(x) = c_1 e^{3x} + c_2 e^{-x}$

Then $y'(x) = 3c_1 e^{3x} - c_2 e^{-x}$ Now $y(0) = 2 \Rightarrow c_1 + c_2 = 2$

and $y'(0) = 14 \Rightarrow 3c_1 - c_2 = 14$

$$\Rightarrow 4c_1 = 16 \Rightarrow c_1 = 4 \text{ and } c_2 = 2 - 4 = -2$$

So, the particular solution of the IVP is

$$y(x) = 4e^{3x} - 2e^{-x}$$

Ex: $y'' + 4\pi y' + 4\pi^2 y = 0$

The characteristic equation is $\lambda^2 + 4\pi\lambda + 4\pi^2 = 0$

$\Rightarrow (\lambda + 2\pi)^2 = 0 \Rightarrow -2\pi$ is a double root of the characteristic

equation $\Rightarrow$ The general solution is $y(x) = e^{-2\pi x} (c_1 + c_2 x)$

Ex: $y'' + 2y' + 5y = 0$, $y(0) = 0$, $y'(0) = 2$

The characteristic equation is:

$$\lambda^2 + 2\lambda + 5 = 0$$

$$\Rightarrow \lambda = \frac{-2 \pm \sqrt{2^2 - 4(5)}}{2} = \frac{-2 \pm 4i}{2} = -1 \pm 2i$$

So, $\lambda_1 = -1 + 2i$ $\lambda_2 = -1 - 2i$

Thus the general solution is $y(x) = e^{-x} (c_1 \cos 2x + c_2 \sin 2x)$

$$y(0) = 0 \Rightarrow c_1(1) + c_2 \times 0 = 0 \Rightarrow \boxed{c_1 = 0}$$

$$y'(x) = -e^{-x} c_2 \sin 2x + e^{-x} 2c_2 \cos 2x$$

Now $y'(0) = 2 \Rightarrow 0 + (1)2c_2 = 2 \Rightarrow \boxed{c_2 = 1}$

So, the solution of the IVP is

$$y(x) = e^{-x} \sin 2x$$

